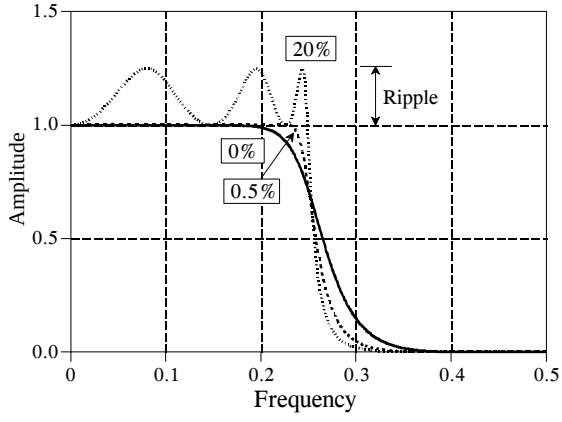

FIGURE 20-1
The Chebyshev response. Chebyshev filters achieve a faster roll-off by allowing ripple in the passband. When the ripple is set to 0%, it is called a *maximally flat* or *Butterworth* filter. Consider using a ripple of 0.5% in your designs; this passband unflatness is so small that it cannot be seen in this graph, but the roll-off is much faster than the Butterworth.

type 2 Chebyshev filters have ripple only in the *stopband*. Type 2 filters are seldom used, and we won't discuss them. There is, however, an important design called the **elliptic filter**, which has ripple in *both* the passband and the stopband. Elliptic filters provide the fastest roll-off for a given number of poles, but are much harder to design. We won't discuss the elliptic filter here, but be aware that it is frequently the first choice of professional filter designers, both in analog electronics and DSP. If you need this level of performance, buy a software package for designing digital filters.

Designing the Filter

You must select four parameters to design a Chebyshev filter: (1) a high-pass or low-pass response, (2) the cutoff frequency, (3) the percent ripple in the passband, and (4) the number of poles. Just what is a *pole*? Here are two answers. If you don't like one, maybe the other will help:

Answer 1- The Laplace transform and z-transform are mathematical ways of breaking an impulse response into sinusoids and decaying exponentials. This is done by expressing the system's characteristics as one complex polynomial divided by another complex polynomial. The roots of the numerator are called *zeros*, while the roots of the denominator are called *poles*. Since poles and zeros can be complex numbers, it is common to say they have a "location" in the complex plane. Elaborate systems have more poles and zeros than simple ones. Recursive filters are designed by first selecting the location of the poles and zeros, and then finding the appropriate recursion coefficients (or analog components). For example, Butterworth filters have poles that lie on a *circle* in the complex plane, while in a Chebyshev filter they lie on an *ellipse*. This is the topic of Chapters 32 and 33.

Answer 2- Poles are containers filled with magic powder. The more poles in a filter, the better the filter works.